

DES MOINES, IA, USA

WMO: 725460

Lat: 41.534N

Lon: 93.653W

Elev: 957

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.4	0.4	-12.9	2.8	-3.2	-8.5	3.6	1.3	29.9	19.8	27.0	22.0	9.4	310	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	93.0	76.1	90.1	75.0	87.4	73.6	78.9	89.1	77.3	86.9	75.6	84.4	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	140.8	85.7	74.3	132.9	83.9	72.7	125.7	82.1	43.2	89.2	41.4	86.7	39.8	84.7	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.6	19.7	-11.3	97.4	5.6	3.2	-15.3	99.7	-18.6	101.6	-21.7	103.4	-25.7	105.7		
			-12.0	81.9	5.4	1.4	-15.8	82.9	-18.9	83.7	-21.9	84.5	-25.8	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	22.6	27.0	39.5	51.9	62.6	72.4	76.6	74.4	67.0	53.9	40.2	28.0	
	DBStd	21.11	12.57	12.57	12.15	9.91	8.39	6.30	5.60	5.37	8.02	9.53	10.57	11.55	
	HDD50	3040	849	647	367	92	7	0	0	0	1	68	324	685	
	HDD65	6065	1314	1064	791	406	148	14	1	3	71	361	744	1148	
	CDD50	3576	0	3	42	149	396	673	824	757	511	189	30	2	
	CDD65	1127	0	0	2	12	73	236	360	295	131	18	0	0	
	CDH74	10212	0	0	18	141	620	2100	3554	2506	1116	153	4	0	
	CDH80	3658	0	0	2	29	160	741	1444	889	363	30	0	0	
Wind	WSAvg	9.7	10.5	10.5	10.9	11.3	10.0	8.9	8.0	7.6	8.5	9.7	10.4	10.0	
Precipitation	PrecAvg	34.6	1.0	1.2	2.2	3.8	4.4	5.0	3.9	4.2	3.3	2.6	1.8	1.4	
	PrecMax	55.9	4.3	2.9	5.8	7.8	10.0	13.5	9.8	13.7	9.5	6.5	6.5	5.4	
	PrecMin	22.1	0.1	0.1	0.2	0.2	1.4	1.0	0.0	0.2	0.6	0.3	0.0	0.0	
	PrecStd	7.2	0.7	0.7	1.3	1.7	1.9	3.0	2.3	3.0	2.0	1.6	1.3	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	63.2	76.4	84.2	89.2	93.3	98.5	95.8	91.9	84.1	73.4	60.0	
		MCWB	45.5	51.1	60.3	64.4	71.5	75.1	77.4	75.5	73.3	67.6	58.8	52.6	
	2%	DB	48.7	56.3	69.6	78.1	84.2	90.2	94.0	91.4	87.9	78.3	66.1	53.3	
		MCWB	41.8	47.4	56.3	61.6	68.5	74.3	77.4	76.2	72.2	64.3	55.8	47.4	
	5%	DB	43.5	49.7	63.7	73.1	80.7	87.4	90.7	87.9	84.2	73.5	60.5	48.0	
		MCWB	38.2	43.1	54.0	58.3	66.0	72.7	76.6	74.5	70.6	62.1	52.1	42.5	
	10%	DB	38.6	44.0	58.1	68.5	77.0	84.5	87.5	84.9	80.6	69.1	56.1	43.3	
		MCWB	34.9	38.8	49.0	55.5	64.0	71.1	74.8	72.8	68.2	59.6	48.6	38.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	54.6	63.6	67.7	74.4	79.2	81.4	80.3	76.7	71.7	61.4	55.4
			MCDB	53.0	61.0	72.8	79.6	85.7	90.1	92.9	90.2	87.9	79.2	68.6	58.5
2%		WB	42.1	48.0	59.4	63.7	71.3	76.9	79.6	78.1	74.4	67.7	57.4	48.8	
		MCDB	47.8	55.1	66.4	74.4	81.6	87.2	90.2	87.6	84.1	73.9	64.7	51.3	
5%		WB	38.6	43.1	54.9	60.4	68.8	75.0	78.0	76.4	72.6	64.4	53.2	42.5	
		MCDB	43.4	49.6	63.1	70.6	77.6	84.3	87.7	85.1	81.3	70.8	59.7	47.6	
10%		WB	35.1	38.8	49.7	57.3	66.2	73.1	76.4	74.6	70.5	60.6	49.1	38.7	
		MCDB	38.3	43.6	58.0	66.2	74.5	81.7	85.3	82.5	77.9	67.9	55.2	43.0	
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.4	18.2	20.0	18.7	17.9	17.8	17.7	19.9	18.8	16.9	15.0	
		MCDBR	22.0	23.2	26.3	27.0	22.5	20.8	20.5	20.9	22.6	23.8	24.0	21.0	
		MCWBR	16.4	16.0	15.5	14.9	11.2	9.8	8.7	9.4	10.5	12.7	15.1	15.5	
	5% WB	MCDBR	20.8	22.3	22.8	23.4	19.9	18.5	18.5	18.1	19.5	19.5	21.4	19.0	
		MCWBR	16.0	16.1	14.1	14.7	11.3	10.4	9.4	9.7	10.4	13.0	15.2	14.9	
Clear-Sky Solar Irradiance	taub	0.290	0.319	0.331	0.388	0.414	0.425	0.440	0.424	0.388	0.344	0.301	0.291		
	taud	2.420	2.341	2.401	2.259	2.255	2.270	2.215	2.274	2.327	2.445	2.504	2.438		
	Ebn at Noon	274	281	292	280	274	270	265	266	268	269	269	264		
	Edh at Noon	25	31	33	42	43	42	44	41	36	28	23	23		
All-Sky Solar Radiation	RadAvg	613	930	1203	1507	1780	1979	1997	1730	1456	980	667	514		
	RadStd	57	61	113	113	141	191	150	113	108	113	56	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+171
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page